Frequentist, Bayesian and Machine Learning Analysis of the Matter–Antimatter Asymmetry in $B^{\pm} \to K^{\pm}K^{+}K^{-}$ Decays at LHCb

Final project for the *Statistics and Data Analysis* course
Academic Year 2025–2026

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[TODO: Giacomo + Marco, write last. 3–5 sentences: dataset (LHCb Open Data, $B^{\pm} \to K^{\pm}K^{+}K^{-}$), the three methodological approaches compared (frequentist MLE/profile likelihood, Bayesian MCMC, ML-assisted selection), the headline result (A_{raw} estimate + consistency across methods), one sentence on what's new/interesting (e.g. goodness-of-fit model comparison, local Dalitz asymmetries). Don't give every number here – see §VIII guidance.]

I. INTRODUCTION

Charge-parity (CP) violation – the small observed asymmetry between the behavior of matter and antimatter – is one of the necessary ingredients for the matter-dominated universe we observe, yet the CP violation built into the Standard Model is far too small to account for it. Studying CP violation in the decays of B mesons, where it is comparatively large and experimentally accessible, is one of the main ways to test the Standard Model's CKM description of the effect and search for physics beyond it. Charmless three-body decays such as $B^{\pm} \to h^{\pm}h^{+}h^{-}$ ($h = K, \pi$) are especially interesting in this respect: unlike a two-body decay, the three-body phase space (the Dalitz plot) allows the raw asymmetry A_{raw} to be studied not only as a single global number, but also *locally*, since interference between intermediate resonances can concentrate CP-violating effects in specific kinematic regions rather than spreading them uniformly [1, 2].

This project uses the LHCb Open Data record for $B^{\pm} \to K^{\pm}K^{+}K^{-}$ [3] to address the research question: *to what extent do frequentist likelihood fits, Bayesian inference, and machine-learning-assisted event selection produce consistent, well-controlled estimates of the raw asymmetry A_{raw} in this decay?* The goal is not to reproduce a competitive LHCb-quality measurement, but to build, on top of a validated public dataset and starting notebook [4], a methodologically rigorous project that puts these three inference frameworks – covered separately in the course – side by side on the exact same data and the exact same physical question. Section II describes the dataset and the preselection shared by all three modules; Secs. III–V present the frequentist, Bayesian, and machine-learning analyses respectively; Sec. VI brings the three together.

II. DATA

A. Dataset and Preselection

The dataset is the CERN Open Data record 4900, "Matter Antimatter Differences – B meson decays to three hadrons" [3]: candidates for $B^{\pm} \to h^{\pm}h^{+}h^{-}$ decays collected by LHCb in 2011 at $\sqrt{s} = 7\,\text{TeV}$, reconstructed with the LHCb software and already passed through trigger and offline selection. The data are distributed as ROOT `TTree`s (`DecayTree`) containing only the most relevant reconstructed branches, split across two files by magnet polarity (magnet up/down), for a combined 8,556,118 candidates. The same record also includes a simplified, detector-response-free phase-space simulation of $B^{\pm} \to K^{\pm}K^{+}K^{-}$; it is not used anywhere in this project (no multi-channel phase-space generation was pursued, to keep the project in scope) and is mentioned here only for completeness.

For each of the three final-state tracks ($H1$, $H2$, $H3$) the analysis reads momentum components (`PX`, `PY`, `PZ`), PID likelihoods (`ProbK`, `ProbPi`), a muon flag (`isMuon`), and the track charge – 18 kinematic/PID branches plus 3 charges. All three analysis modules (frequentist, Bayesian, machine learning) share the same preselection, agreed at the start of the project so results stay comparable: a kaon-PID cut and a muon veto applied to all three tracks,

$$\texttt{H*_ProbPi} < 0.5 \;\wedge\; \texttt{H*_ProbK} > 0.5 \;\wedge\; \neg\texttt{H*_isMuon},$$

which retains 23,895 of the 8,556,118 read candidates (0.28%) – the PID requirement, applied on all three tracks simultaneously, is by far the dominant reduction, consistent with selecting a comparatively pure KKK final state out of a much larger, PID-unconstrained sample.

While implementing the Dalitz-plot analysis (Sec. V E), it was found that in this dataset $H1$ is, by construction, always the track of opposite charge to the other two: only the charge combinations $(+1, -1, -1)$ and $(-1, +1, +1)$ occur for $(H1, H2, H3)$. Equivalently, $H2$ and $H3$ are always the same-sign, interchangeable pair

(the two kaons sharing the sign of the B), while $H1$ is the unique kaon of opposite sign – a labeling convention that matters when constructing the Dalitz-plot variables, which are the invariant masses of the two opposite-sign pairs, $m^2(H1, H2)$ and $m^2(H1, H3)$.

B. Invariant Mass Reconstruction

The three final-state tracks are assigned the charged-kaon mass hypothesis ($M_K = 493.677$ MeV) regardless of their PID likelihoods, and their four-momenta are summed to reconstruct the B candidate, $m(KKK) = \sqrt{(\sum_i E_i)^2 - |\sum_i \vec{p}_i|^2}$. A first look at the full reconstructed mass range showed a rise in combinatorial background below ∼5150 MeV, consistent with a partially-reconstructed contribution that a single exponential background shape cannot describe; the fit window used throughout this project – by both the frequentist and Bayesian modules, and to build the ML weak labels (Sec. V A) – was therefore set to $[5150, 5450]$ MeV. This leaves 17,839 preselected candidates in the fit window: 9358 with B-charge $+1$ and 8481 with B-charge -1, split 10,500 (magnet down) / 7339 (magnet up).

C. Data Exploration

Figure I shows the reconstructed mass spectrum in the fit window, split by B charge (left) and by magnet polarity (right). The spectrum peaks at $m \approx 5284$ MeV with a visible signal excess over a falling combinatorial background, and its shape is qualitatively similar for the two charges and the two magnet polarities – as expected, since a large raw asymmetry would already be visible by eye at this level. The simple charge-counting asymmetry on the full fit-window sample, $A_{\text{count}} = (N^- - N^+)/(N^- + N^+) = -0.049$, is numerically close to every fit-based estimate obtained later in the project (Secs. IV C, V F). Since A_{count} mixes signal and background, this closeness is itself informative: with a charge-symmetric background one would expect $A_{\text{count}} \approx A_{\text{raw}}\, N_s/(N_s + N_b) \approx -0.043$, so the data suggest a background asymmetry of similar sign and size to the signal one (see the sideband check of Sec. IV F). For the ML module, the corresponding weak-label class balance (15,119 signal-like vs. 1217 background-like events, Sec. V A) reflects the same high-purity signal region, seen here from the sideband side rather than the counting side.

III. FREQUENTIST ANALYSIS

As of this draft, branch `freq` (Marco's) still holds only the pristine official starting notebook plus a TODO checklist – none of the numerical results below have been produced yet. The subsections are laid out now, following the shared plan (`Docs/Proposta_progetto_LHCb_gruppo.tex`, §"Estensione frequentista"), so results can be dropped in directly once available, with the same structure and cross-references as the parallel Bayesian analysis (Sec. IV).

A. Mass Model and Fit Strategy

The frequentist fit reuses the same physical mass model as the Bayesian analysis (Sec. IV A), $f(m) = N_s\, S(m;\theta_s) + N_b\, B(m;\theta_b)$, estimated by unbinned maximum likelihood (`iminuit`, `iminuit.cost.UnbinnedNLL`) rather than by MCMC. The plan is to start from the same minimal shapes as the Bayesian baseline (Gaussian signal, exponential background) and increase model complexity only if the residuals justify it – mirroring the Bayesian goodness-of-fit finding (Sec. IV E) that the minimal model is not fully adequate ($\chi^2/\text{ndof} = 2.36$) and that a Crystal Ball with floated tail parameters improves it. **[TODO: Marco – fill in once the mass fit is run: final signal/background shapes chosen, fitted yields N_s, N_b, and any residual-based justification for departing from the minimal model.]**

B. Separate vs. Simultaneous Fit

Two parametrizations are planned: a *separate* fit of B^+ and B^- independently, giving N_s^+ and N_s^- from which A_{raw} is computed post hoc, and a *simultaneous* fit using the shared parametrization $N_s^+ = N_s(1 - A)/2$, $N_s^- = N_s(1 + A)/2$ (the same parametrization already used on the Bayesian side, Sec. IV A) that estimates A directly as a fit parameter. Mass, resolution, and background-shape parameters are treated as nuisance parameters, profiled out when reporting the uncertainty on A. **[TODO: Marco – report A_{raw} from both parametrizations (should agree if the model is well-specified) and flag any disagreement as a modeling issue rather than a physics result.]**

C. Profile Likelihood and Confidence Interval

The profile likelihood ratio,

$$q(A) = -2 \log \frac{L(A, \hat{\hat{\theta}})}{L(\hat{A}, \hat{\theta})},$$

is used to derive a confidence interval on A via Wilks' theorem, and to test the null hypothesis $A = 0$ – the frequentist counterpart of the Bayesian Savage-Dickey test of Sec. IV D. **[TODO: Marco – report the MLE $\hat{A}$, the 68%/95% profile-likelihood confidence intervals, and the p-value/significance against $A = 0$; compare directly against the Bayesian credible interval of Sec. IV C ($A_{\text{raw}} = -0.0497$, 68% CI $[-0.058, -0.041]$) once available.]**

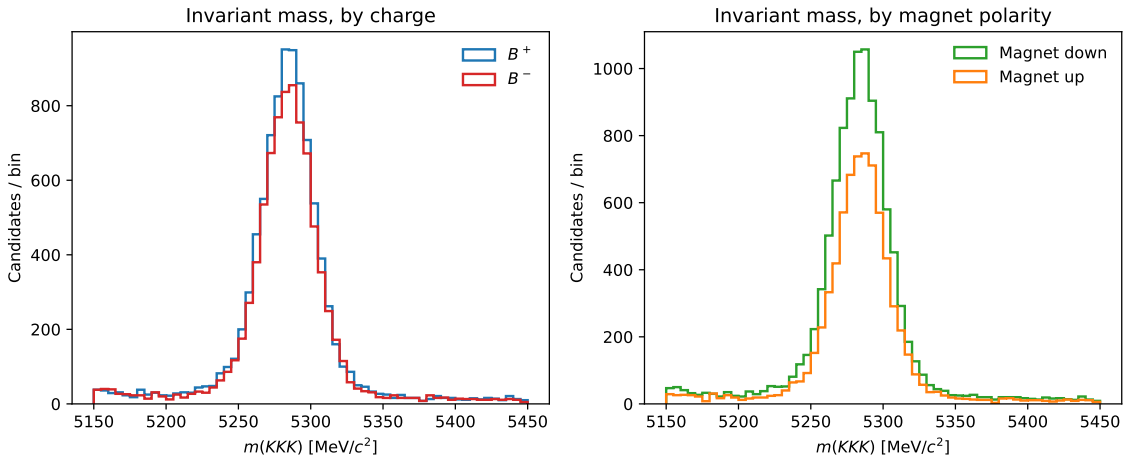


FIG. I. Reconstructed $B \to KKK$ invariant mass in the fit window $[5150, 5450]$ MeV, split by B charge (left) and by magnet polarity (right). Both splits show a consistent peak at $m \approx 5284$ MeV over a falling combinatorial background.

D. Systematic Uncertainties

[TODO: Marco – systematic-uncertainty study, analogous in spirit to the Bayesian systematic-variation table of Sec. IV F (Table III: mass window, background shape, signal-model choice), so the two can be placed side by side in Sec. VI and in Appendix A.]

IV. BAYESIAN ANALYSIS

A. Model and Priors

The Bayesian analysis reuses the same physical mass model as the frequentist fit (Sec. III): a signal-plus-background mixture

$$f(m) = N_s\, S(m;\mu,\sigma) + N_b\, B(m;\lambda),$$

with S a truncated Gaussian and B a truncated exponential, both normalized on the fit window $[5150, 5450]$ MeV (Sec. II B). The signal yield is split between the two B-meson charges through a simultaneous parametrization, $N_s^+ = N_s(1-A)/2$ and $N_s^- = N_s(1+A)/2$, so that A is a fit parameter rather than being derived from two independent yields; the background is assumed to be shared equally between charges, $N_b^\pm = N_b/2$. The likelihood is the extended unbinned likelihood of both charge samples.

The full parameter vector is $\theta = (N_s, N_b, A, \mu, \sigma, \lambda)$, with independent bounded-uniform priors, following the convention of the course's MCMC exercises: $N_s, N_b \in (0, 5N)$ with $N = 17839$ the number of selected events, $A \in (-1, 1)$, $\mu \in (5260, 5310)$ MeV, $\sigma \in (3, 40)$ MeV, $\lambda \in (10^{-5}, 0.05)$ MeV^{-1}. Bounds and the walker starting point were set from a first look at the preselected mass spectrum (peak at $\mu \approx 5283$ MeV).

B. MCMC Sampling and Convergence Diagnostics

The posterior was sampled with `emcee` using an ensemble of 32 walkers initialized in a narrow ball around the starting point, run for 9000 steps with the first 1500 ($\sim$1/6) discarded as burn-in, leaving 2.4×10^5 samples. The mean acceptance fraction is 0.52. Convergence was checked through the integrated autocorrelation time τ of each parameter, $\tau \in [58, 79]$ steps: the post-burn-in chain is $N/\tau \geq 95$ autocorrelation times long for every parameter, above the usual threshold of 50 for a reliable τ estimate, and corresponds to $\gtrsim 3000$ effectively independent samples per parameter.

C. Posterior Results and Credible Intervals

The marginal posterior on the raw asymmetry gives (median and central intervals)

$$A_{\text{raw}} = -0.0497,$$
$$68\%\ \text{CI}\ [-0.058, -0.041], \qquad 95\%\ \text{CI}\ [-0.066, -0.033].$$

The posterior is very close to Gaussian, with standard deviation 0.0083. The result has the same sign and order of magnitude as the published $A_{CP}(KKK) = -0.037 \pm 0.002\,(\text{stat}) \pm 0.002\,(\text{syst}) \pm 0.003\,(J/\psi K)$ [1, 2]; this is a sanity check only, since A_{raw} is not corrected for detection or production asymmetries and is therefore not directly comparable to A_{CP} (Sec. VI). The fitted yields are $N_s = 15281 \pm 130$ and $N_b = 2555 \pm 65$, with $\mu = 5284.41 \pm 0.16$ MeV and $\sigma = 17.95 \pm 0.13$ MeV (Fig. II).

D. Bayes Factor: $A = 0$ vs A Free

The course's worked example (Week 4) computes a Bayes factor by numerical integration of the evidence on

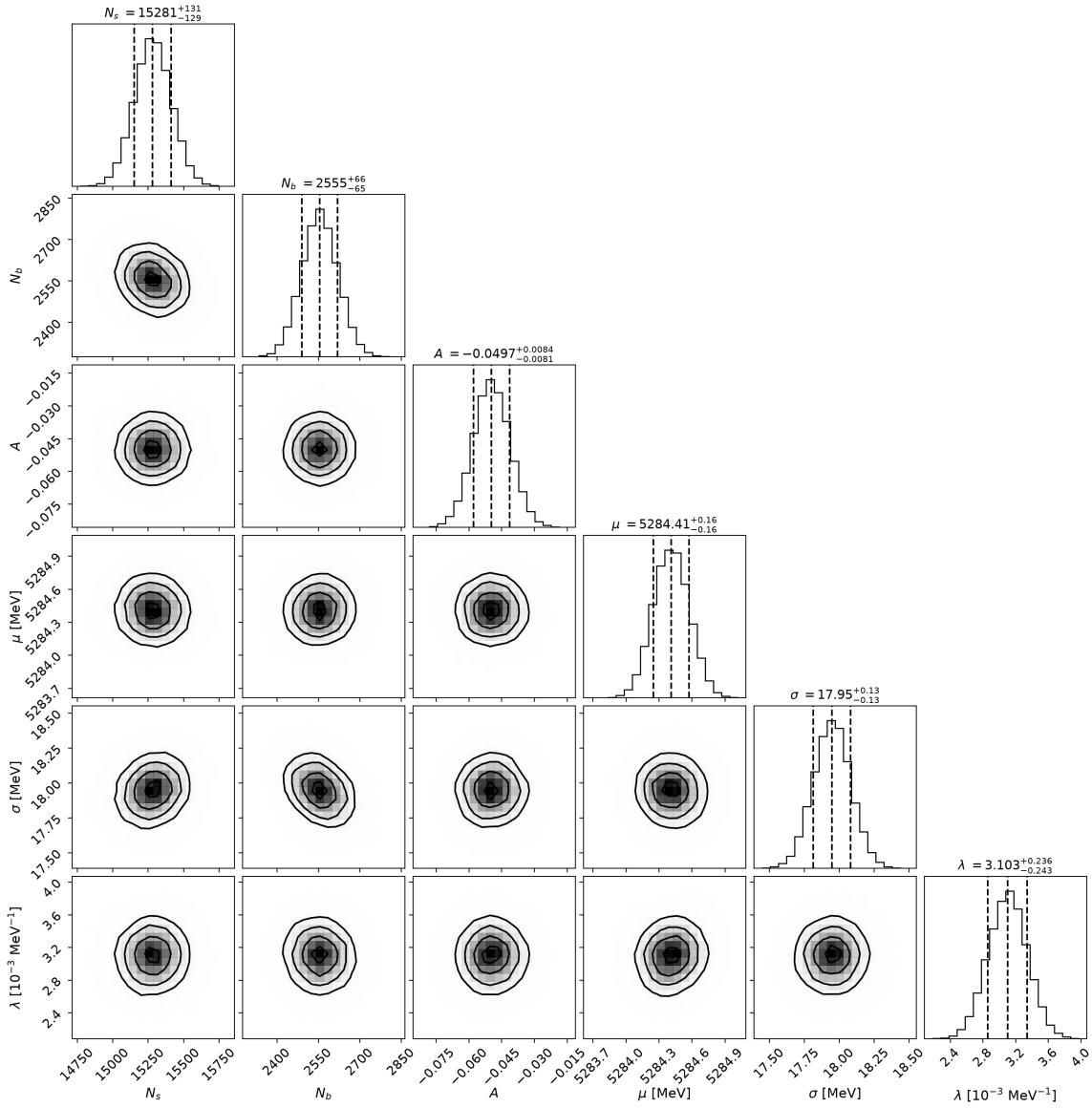


FIG. II. Corner plot of the 6-parameter posterior, nominal fit. Diagonal: marginal posteriors, with dashed lines at the 16th, 50th and 84th percentiles (also given in the titles). Off-diagonal: 2D marginals, with `corner`'s default contours at the $0.5, 1, 1.5, 2\sigma$ levels of a 2D Gaussian. A is essentially uncorrelated with the other parameters; the only visible correlations are mild ones between N_b and N_s and between N_b and σ.

a 1D grid, which does not scale to a 6-parameter posterior. Since H_0 ($A = 0$) is nested in H_1 (A free) and the priors on the other parameters are shared, we use the Savage-Dickey density ratio,

$$BF_{01} = \frac{p(A = 0 \mid \text{data})}{\pi(A = 0)}, \qquad \pi(A = 0) = \tfrac{1}{2}.$$

$A = 0$ lies 6.0 posterior standard deviations from the median, deep in a tail where the chain contains essentially no samples. Estimating $p(A = 0 \mid \text{data})$ with a Gaussian KDE on the A chain is therefore unreliable: at the Silverman bandwidth it gives $BF_{01} \approx 3 \times 10^{-59}$, and rescaling the bandwidth by 0.5–3 moves the result between 10^{-228} and 10^{-9}, since the KDE value at $A = 0$ only reflects the kernel tails of the few most extreme samples. Because the marginal posterior of A is very close to Gaussian, a more meaningful estimate uses its Gaussian approximation, giving $BF_{01} \approx 1.3 \times 10^{-6}$ ($\ln BF_{10} \approx 13.5$): decisive evidence against $A = 0$ on the Jeffreys scale, as is every KDE value above. As for any Bayes factor of a point null against a free parameter, the value scales with the prior width (Lindley's paradox): with the narrower prior $A \in (-0.3, 0.3)$ of Sec. IV F, $\pi(A = 0)$ grows to $1/0.6$ and BF_{01} drops by a factor 3.3, which leaves the conclusion unchanged.

E. Goodness-of-Fit and Signal Model Comparison

A binned comparison of data and posterior-median model ($\chi^2 = \sum(\text{obs} - \text{exp})^2/\text{exp}$ over 30 mass bins per charge) shows that the minimal Gaussian signal model does not fully describe the data: $\chi^2/\text{ndof} = 127.4/54 = 2.36$, with $\chi^2 = 80.4$ for B^+ and 47.1 for B^-. The largest pulls sit on the two flanks of the peak, about $\pm 3\sigma$ from μ (Fig. III): the observed signal shape has heavier tails than a Gaussian. Table I compares three signal shapes. A Crystal Ball with tail parameters fixed to typical values ($\beta = 1.5$, $m = 3$) makes the fit worse, not better. Letting β and m float (an 8-parameter fit) improves χ^2/ndof to 2.00, with $\beta = 1.83$ and $m = 12.4$, i.e. a tail specific to this dataset. Across the three models A_{raw} changes by at most 0.0015, a fifth of its statistical uncertainty, so the asymmetry does not depend on the signal shape even though none of the models reaches $\chi^2/\text{ndof} \approx 1$.

Signal model	A_{raw}	68% CI	χ^2/ndof
Gaussian	-0.0497	$[-0.058, -0.041]$	2.36
Crystal Ball (fixed β, m)	-0.0512	$[-0.059, -0.043]$	4.50
Crystal Ball (free β, m)	-0.0504	$[-0.059, -0.042]$	2.00

TABLE I. Signal-model comparison. A_{raw} is stable; goodness-of-fit is not, and improves only once the Crystal Ball tail parameters are fitted rather than fixed.

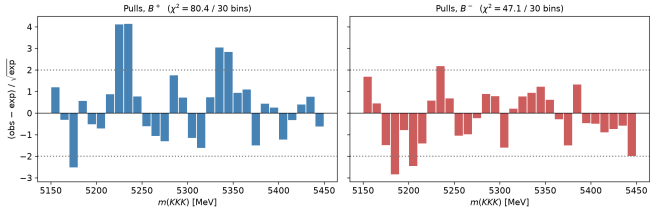


FIG. III. Binned pulls, $(\text{obs} - \text{exp})/\sqrt{\text{exp}}$, of the nominal Gaussian-signal model, separately for B^+ and B^-. The largest positive pulls lie on the flanks of the peak (~5230 and ~5340 MeV, about $\pm 3\sigma$ from μ), the signature of non-Gaussian tails; they drive $\chi^2/\text{ndof} = 2.36$ and motivate the Crystal Ball comparison of Table I.

F. Robustness Checks: Mass Window, Background Model, Prior, Magnet Polarity

The nominal fit was repeated under three alternative modeling choices – a narrower window ($[5200, 5450]$ MeV), a window without the sparse high-mass tail ($[5150, 5400]$ MeV), and a linear instead of exponential background – and with a narrower prior on the asymmetry, $A \in (-0.3, 0.3)$, as a prior-sensitivity check. All four use the nominal chain length (9000 steps, 32 walkers). They give $A_{\text{raw}} \in [-0.0498, -0.0495]$ (Fig. IV, Table III): the spread, 3×10^{-4}, is about 4% of the statistical uncertainty and only about twice the Monte Carlo error on the posterior median ($\sim 1.4 \times 10^{-4}$), so no sensitivity to these choices is detected.

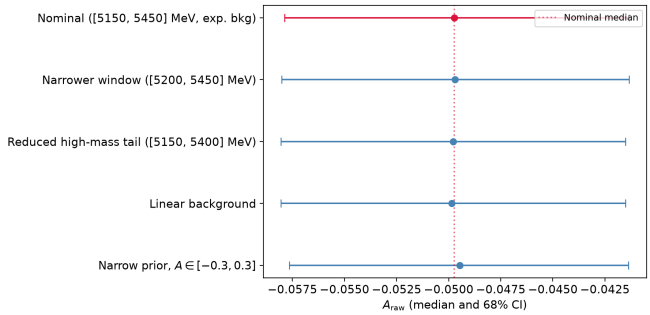


FIG. IV. Posterior median (point) and 68% CI (bar) of A_{raw} for the nominal fit and the four variations of Table III. The five results are indistinguishable on the scale of the statistical uncertainty.

As an experimental cross-check, the fit was repeated separately on the two magnet polarities, whose left-right detection asymmetries are opposite and partially cancel in the combined sample: $A_{\text{raw}}(\text{down}) = -0.055$ ($[-0.065, -0.044]$, $n = 10500$) and $A_{\text{raw}}(\text{up}) = -0.043$ ($[-0.056, -0.030]$, $n = 7339$). They differ by 0.7σ, so there is no indication that the asymmetry is an artifact of one polarity (Fig. V).

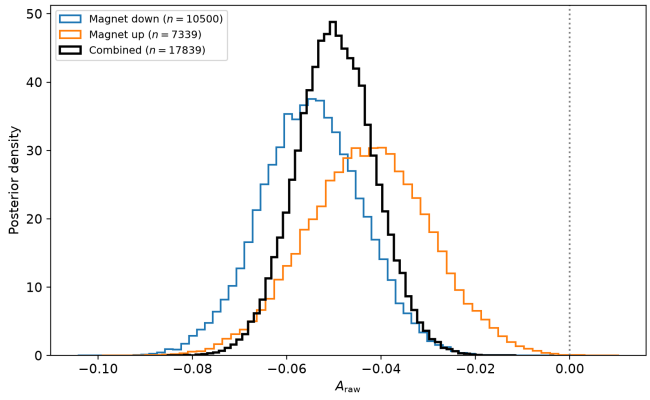


FIG. V. Posterior on A_{raw} for the magnet-down-only, magnet-up-only, and combined fits. The two single-polarity posteriors differ by 0.7σ.

The assumption of a charge-symmetric background, $N_b^{\pm} = N_b/2$, was tested on the sidebands ($|m - \mu| > 4\sigma$, where the expected signal contamination is negligible): their counting asymmetry is $A_{\text{sideband}} = -0.047 \pm 0.029$ ($n = 1217$), 1.6σ from zero and equally compatible with the signal asymmetry. The same picture emerges from the full fit-window count (Sec. II C), which is closer to A_{raw} than a symmetric background would allow. The check is therefore inconclusive rather than reassuring. The definitive test is a fit with independent background

yields N_b^+, N_b^-, as in the frequentist joint fit (Sec. III). A simple estimate suggests the impact on $A_{\rm raw}$ is small: if the background carried $A_b = A_{\rm sideband}$, attributing the excess $A_b N_b$ to the background rather than to the signal would give $(N^- - N^+ - A_b N_b)/N_s \approx -0.0495$.

G. Toy-MC Coverage Validation

To check whether the credible intervals have the claimed frequentist coverage, 500 pseudo-datasets of the same size as the data were generated from the fitted model (parametric bootstrap, with the posterior-median θ as ground truth) and each was refit with a 3500-step chain. The fraction of replicas whose interval contains the injected A is 0.664 for the 68% CI (0.8σ below nominal, binomial uncertainty 0.021) and 0.962 for the 95% CI (1.2σ above nominal): the intervals have correct coverage. This is a closure test of the inference procedure: since the toys are generated from the same model that is fitted, it does not probe the model mismatch of Sec. IV E, which is instead addressed by the signal-model and robustness checks.

H. Local Dalitz-Plot Asymmetry (Fixed Binning)

To test whether the asymmetry is uniform over phase space or concentrated in specific regions, each candidate was placed in the Dalitz plane using the two opposite-sign K^+K^- invariant masses $m^2(H1,H2)$ and $m^2(H1,H3)$ (Sec. II A), sorted into $m^2_{\rm lo}$ and $m^2_{\rm hi}$ to remove the $H2/H3$ labeling ambiguity, as in the official starting notebook. The plane was divided into an 8×8 grid. In each bin with at least 20 candidates (29 of 64), the local asymmetry $A_i = (N_i^- - N_i^+)/N_i$, with binomial uncertainty $\sigma(A_i) = \sqrt{(1-A_i^2)/N_i}$, was compared with the global counting asymmetry $A_{\rm glob} = -0.049$ through $S_i = (A_i - A_{\rm glob})/\sigma(A_i)$.

Three of the 29 bins have $|S_i| > 3$, all with A_i more negative than the global value; about 0.08 would be expected if the asymmetry were uniform, with a probability of 7×10^{-5} for three or more. A global test of homogeneity, $\chi^2 = \sum_i S_i^2 = 85.6$ for 28 degrees of freedom ($p \approx 10^{-7}$), confirms that the local asymmetry is not uniform. The most negative bins lie at low $m^2_{\rm lo}$ (Fig. VI). A test against zero instead of $A_{\rm glob}$, as in the starting notebook, would partly reflect the global asymmetry itself. Restricting the sample to the signal window ($|m-\mu| < 2\sigma$, n = 15,119, ≈96% signal purity from the fitted model) gives the same picture – 29 populated bins, three with $|S_i| > 3$, $\chi^2 = 95.5/28$ – so the structure is not produced by the combinatorial background. Two caveats apply: the binning is coarse and was not optimized, and $\sigma(A_i)$ ignores the background fluctuations, which a proper sWeight-based background subtraction would include (Sec. VII). Section V E revisits the same question with an adaptive, data-driven binning (K-means).

V. MACHINE LEARNING MODULE

As a complementary, purely data-driven approach to the signal/background separation used in the mass fits (Secs. III–IV), a supervised classifier was trained on weak labels derived from the mass sidebands and applied to study whether an ML-assisted event selection biases $A_{\rm raw}$, and whether it can resolve local Dalitz-plot structure that a fixed binning would miss.

A. Weak Labels and Feature Set

Weak labels were built from the same mass model as the Bayesian fit (Sec. IV A, $\mu = 5284.4$, $\sigma = 17.95$ MeV), without ever using the reconstructed B mass itself as an input feature, to avoid mass sculpting. Events with $|m-\mu| < 2\sigma$ (i.e. $m \in [5248.5, 5320.3]$ MeV) are labeled signal-like; events with $|m-\mu| > 4\sigma$ (outside $[5212.6, 5356.2]$ MeV, within the fit window) are labeled background-like; the intermediate 2–4σ gap is left unlabeled and excluded, to keep the two classes cleanly separated. This gives $n_{\rm labeled} = 16336$ events, of which 15119 are signal-like and only 1217 background-like (imbalance ratio 0.080) – a severe class imbalance, handled downstream with `class_weight='balanced'` for the scikit-learn models, a sample-weight approximation for GBM (which has no native `class_weight` support), and explicit class weights in the Keras loss for the FFNN.

The feature set consists of the 6 per-track kinematic/PID variables (`PX`, `PY`, `PZ`, `ProbK`, `ProbPi`, `IPChi2`) for each of the 3 final-state tracks (18 features), plus two B-candidate-level variables found unused in the shared preselection and worth checking as classifier inputs: `B_FlightDistance` and `B_VertexChi2`, both standard LHCb combinatorial-background discriminants (a genuine B flies further before decaying and has a better-reconstructed vertex than a random track combination). An exploratory check confirmed both separate the weak-label classes significantly (Kolmogorov-Smirnov $p < 10^{-35}$ for each) and are not redundant with the existing features (Spearman $|\rho| < 0.52$ with every one of the 18), so both were adopted, for 20 features total. An exploratory PCA on the standardized features shows 15 of 20 components are needed to reach 90% explained variance – the features carry mostly independent information, so no dimensionality reduction is applied to the models used downstream (PCA is kept as a diagnostic only). The labeled sample was split 60/20/20 into train/validation/test, stratified on the label, with the scaler fit on the training set only.

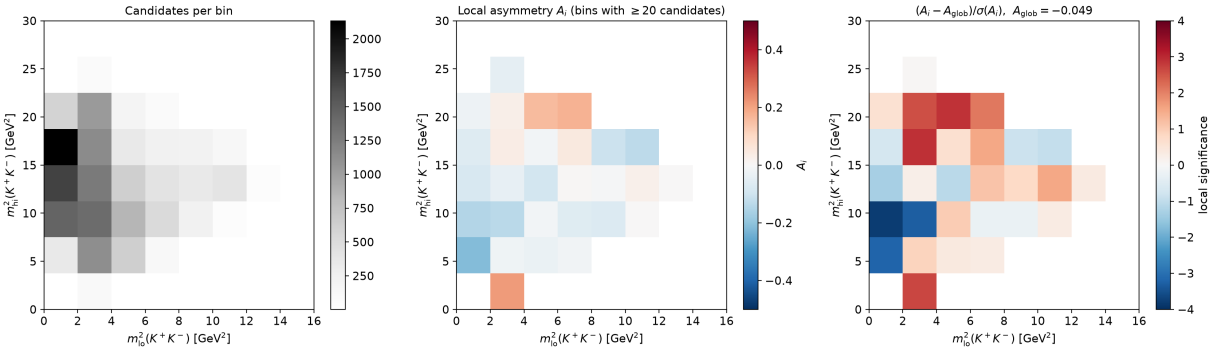


FIG. VI. Fixed 8×8 Dalitz-plot binning, all candidates in the fit window: number of candidates per bin (left), local asymmetry A_i for bins with ≥ 20 candidates (centre), and local significance with respect to the global asymmetry, $(A_i - A_{\rm glob})/\sigma(A_i)$ (right). The bins with $|S_i| > 3$ (dark blue) lie at low $m^2_{\rm lo}$.

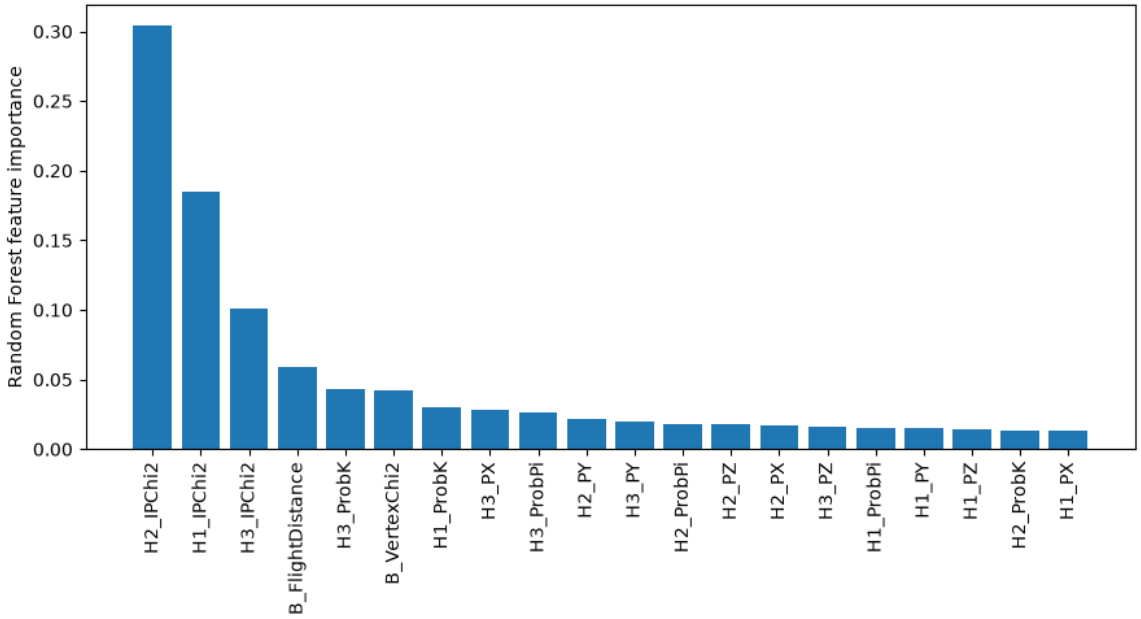


FIG. VII. Random Forest feature importance (Gini), all 20 features. The three `IPChi2` variables dominate, but `B_FlightDistance` (4th) and `B_VertexChi2` (6th) both rank above most of the per-track PID/kinematic variables – confirming, on the trained model, the exploratory finding that motivated adopting them (Sec. V A).

B. Model Comparison

Five classifier families were trained: logistic regression (linear baseline), a single decision tree, Random Forest, Gradient Boosting, and a feed-forward neural network (two hidden layers, 32–16 units, dropout 0.2). Table II summarizes 5-fold cross-validated AUC on the training set (not computed for the FFNN, evaluated on the held-out validation set instead) and AUC on the untouched test set.

Both ensembles (Random Forest, GBM) improve test AUC by $\sim$0.05–0.06 over the single decision tree and by $\sim$0.08 over plain logistic regression, consistent with the mild non-linearity expected in a PID/kinematics-based classifier. Adding the two B-level vertex-quality features (Sec. V A) improved every model's AUC over the 18-feature baseline, most visibly for the linear models (+0.03 for logistic regression) and more modestly for the tree ensembles (+0.001–0.002) – consistent with the linear model previously being unable to exploit the (nonlinear) discriminating power that the trees already partly captured indirectly. Random Forest was selected as the reference model for all downstream analyses (Secs. V D–V F) based on its best CV AUC, essentially tied with GBM on test AUC.

Model	CV AUC (train)	Test AUC
Logistic Regression	0.7273 ± 0.0180	0.7326
Logistic Regression (L1)	–	0.7333
Decision Tree	0.7470 ± 0.0213	0.7575
Random Forest	0.8136 ± 0.0189	**0.8127**
Gradient Boosting	0.8154 ± 0.0192	0.8127
FFNN	–	0.7681

TABLE II. Cross-validated and test-set AUC for all six models, 20-feature classifier. The two tree ensembles clearly outperform the linear and single-tree baselines; Random Forest and GBM are statistically tied on test AUC, Random Forest is used as the reference model for Secs. V D–V F.

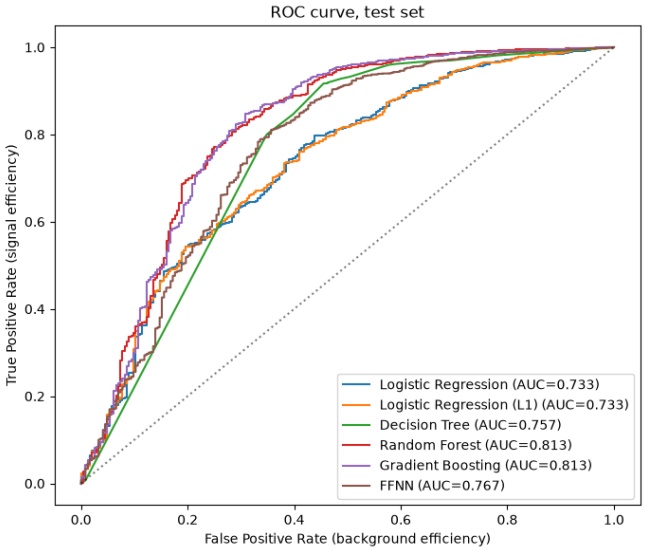


FIG. VIII. ROC curves, test set, all six models. The two tree ensembles (red, purple) dominate the other four across essentially the whole curve.

C. Regularization, Bias-Variance, and Calibration

An $L1/L2$ regularization sweep on the logistic regression (log-spaced grid on $C = 1/\alpha$, with CV) finds essentially identical optimal AUC for both penalties ($C_{L1} = 0.024$, AUC $= 0.7280$; $C_{L2} = 45.2$, AUC $= 0.7273$), with 3 of the 20 features driven to exactly zero by the $L1$ penalty (only 1 with 18 features) – the linear model is still not meaningfully overfitting and gains little from aggressive regularization, consistent with its comparatively low AUC ceiling in Table II, though the larger feature set gives $L1$ slightly more to prune.

A bias-variance complexity curve was built for GBM using `staged_predict` (train/test AUC as a function of the number of boosting stages). The train/test AUC gap grows from 0.014 at 20 trees to 0.033 at the final iteration – mild but measurable overfitting as complexity increases, though the gap stays small relative to the overall AUC scale (∼0.81).

Calibration was assessed via the Brier score for all six models (Table II companion metric): Logistic Regression 0.208, Logistic Regression (L1) 0.212, Decision Tree 0.153, Random Forest 0.132, Gradient Boosting 0.125, FFNN 0.071. Notably the FFNN, despite the lowest AUC of the four non-linear models, has by far the best-calibrated raw score – a model can rank events well (high AUC) without its output being a trustworthy probability, and vice versa. Applying isotonic recalibration (`CalibratedClassifierCV`) to the reference Random Forest model reduces its Brier score from 0.132 to 0.053 (a 60% reduction): the RF score is a good *ranking* but, uncalibrated, a poor *probability* estimate.

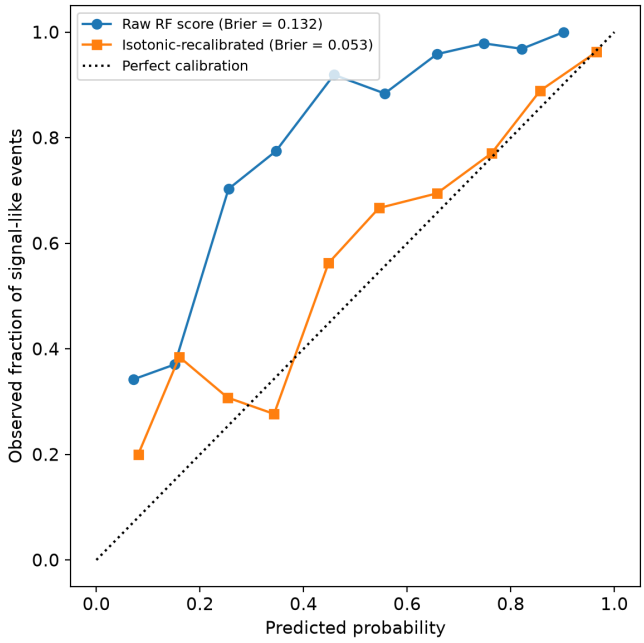


FIG. IX. Reliability diagram for the reference Random Forest, raw score vs. isotonic-recalibrated score. The recalibrated curve tracks the diagonal (perfect calibration) far more closely.

D. Uncertainty Quantification

Two independent estimates of epistemic (model) uncertainty were computed per event: MC Dropout on the FFNN (dropout kept active at inference, 300 repeated forward passes, uncertainty = std across passes) and ensemble-variance on the Random Forest (uncertainty = std of the individual trees' predictions, following the bagging-uncertainty prescription from Lecture 18). The two give mean uncertainties of 0.0112 (MC Dropout) and 0.185 (RF ensemble) – a factor ∼16.5 apart in absolute scale.

Two checks were run to understand the discrepancy. First, whether the class imbalance (Sec. V A) inflates RF tree-to-tree variance on the rare background-like class: the mean RF std is 0.196 on background-like test events

vs. 0.185 on signal-like – only a ∼6% relative difference, too small to explain a 16.5× gap. The same comparison for MC Dropout shows a much larger relative effect (0.0224 background vs. 0.0103 signal, ∼2.2×), suggesting the FFNN's epistemic uncertainty is more sensitive to the scarcity of background-like training examples than the RF's is. Second, whether the two methods at least agree on *which* events are most uncertain, independent of absolute scale: the Spearman correlation between per-event MC-Dropout and RF-ensemble uncertainty is $\rho = 0.368$ ($p = 2.1 \times 10^{-105}$) – highly significant and moderate, meaning the two epistemic-uncertainty estimates agree on relative event-level ranking more than with the 18-feature set ($\rho = 0.24$ previously), but are still far from interchangeable.

E. K-means on the Dalitz Plot

As an adaptive alternative to the fixed Dalitz-plot binning, K-means clustering ($K = 8$, matching the fixed-binning granularity for direct comparison) was run on the standardized kinematic features. Repeating the clustering with different random seeds gives an adjusted Rand index of 0.996 between runs, indicating a highly stable partition. Cluster sizes range from $n = 1216$ to $n = 4007$, and the local raw asymmetry A_{local} per cluster ranges from -0.139 to $+0.035$. Of the 8 clusters, exactly one shows a local deviation from zero exceeding 3σ: cluster 2 ($n = 3341$, $A_{\text{local}} = -0.139$). Restricting that cluster to the high-purity signal window ($|m - \mu| < 2\sigma$, $n_{\text{purified}} = 2880$, ∼85% purity) gives $A_{\text{local}} = -0.147$ – the asymmetry survives, and if anything strengthens slightly, after purification, disfavoring background contamination as its origin.

$K = 8$ was chosen for direct comparability with the fixed binning, not from a clustering-quality criterion – checked after the fact via elbow (inertia) and silhouette score over $K = 2$–12. The global silhouette optimum is $K = 3$ (0.467), but $K = 8$ sits at a local silhouette maximum (0.410, higher than the neighboring $K = 6, 7$ and $K = 9$–12), so it is not an arbitrary choice, just not the statistically optimal one – a coarser, $K = 3$ partition would likely be less directly comparable to the 8×8 fixed binning of Sec. IV H.

F. Stability of A_{raw} under ML Selection

The reference Random Forest score was applied to *all* events in the fit window (not only the labeled subsample used for training), and the counting asymmetry A_{raw} was recomputed after cutting at score thresholds from 0.1 to 0.9, with non-parametric bootstrap uncertainties (resampling the full selected sample and repeating cut+count) at each threshold. Across thresholds 0.1–0.7, A_{raw} stays in a narrow band, $[-0.054, -0.049]$, consistent within uncertainty with the nominal Bayesian fit result (-0.0497, Sec. IV C) and with bootstrap errors that are small and stable (0.007–0.010). At threshold 0.8, A_{raw} drifts to -0.014, but the bootstrap uncertainty there is also much larger (0.026), so the drift is only $\sim 1.4\sigma$ from nominal – suggestive but not significant. At threshold 0.9 the selection becomes fully degenerate ($A_{\text{raw}} = -1$, every surviving event has the same charge, bootstrap uncertainty undefined) – with 20 features the classifier is even more aggressive at extreme thresholds than with 18, leaving too few events to say anything statistically. Over the range where the selection retains a usable sample size, the ML-based event selection does not introduce a detectable bias on A_{raw}.

The threshold sweep above is simple post-cut counting, previously flagged as a limitation (Sec. VII) since it does not test whether the ML selection biases a genuine *fit*-based estimate rather than just a count. This was addressed directly: the Bayesian mass model (Sec. IV A) was refit – same parametrization, same MCMC settings (32 walkers, 9000 steps) – on the RF-selected subsample alone (score > 0.5, $n = 15204$), instead of counting. The result, $A_{\text{raw}} = -0.0493$ (68% CI $[-0.058, -0.041]$), is consistent with both the nominal full-sample fit (-0.0497, Sec. IV C) and the simple-counting estimate on the same subsample (-0.0499) – the ML selection does not bias a proper mass-fit estimate either, closing this limitation on the Bayesian side (the analogous frequentist-side comparison remains blocked on Sec. III).

As with the Bayesian fit (Sec. IV F), the ML-selected sample was also cross-checked by magnet polarity, to test whether the RF cut interacts with the detector's left-right instrumental asymmetry (Sec. IV F) rather than tracking a genuine physical effect. At the representative RF-score threshold of 0.5: $A_{\text{raw}}(\text{down}) = -0.062 \pm 0.011$ ($n = 8871$) and $A_{\text{raw}}(\text{up}) = -0.032 \pm 0.013$ ($n = 6333$). The two are individually somewhat more separated than in the Bayesian check ($\sim 1.8\sigma$ apart here, vs. fully overlapping there), but follow the same sign pattern (down more negative than up) and their 68% intervals still overlap – consistent with the same underlying physical asymmetry seen through a smaller, ML-selected sample, not with a polarity-dependent artifact of the classifier.

VI. COMPARISON AND DISCUSSION

[TODO: Shared, written last once §III exists – the one section that genuinely can't be finished without Marco. Headline comparison table: A_{raw} (counting, frequentist, Bayesian, ML-assisted) with uncertainties, side by side. Comparison with the published A_{CP}(KKK) value from Ref. [1] and discussion of why $A_{\text{raw}} \neq A_{CP}$. Local Dalitz-plot asymmetry: fixed binning vs K-means, and whether the frequentist local significance study (if Marco does one) agrees.]

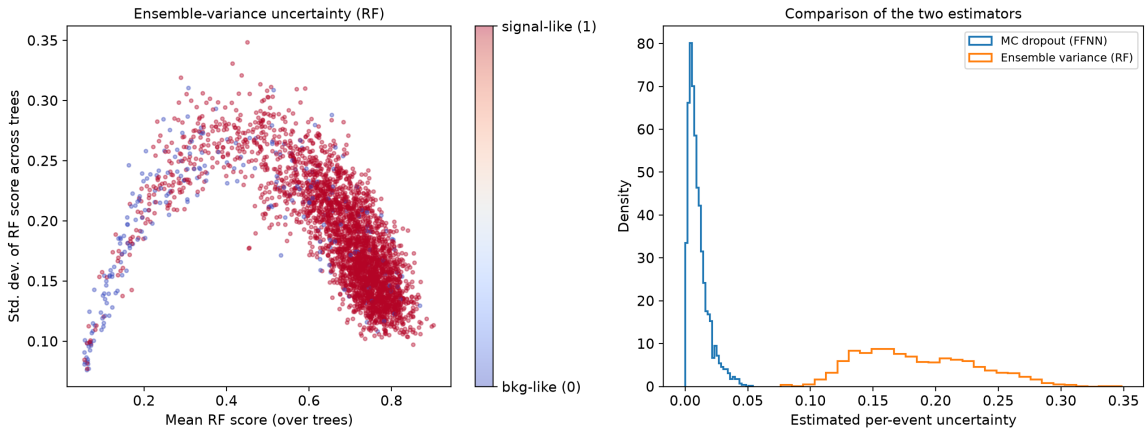


FIG. X. Left: RF ensemble-variance uncertainty vs. mean RF score, per test event (colour = true label). Right: distributions of the two epistemic-uncertainty estimates, MC Dropout (FFNN) and ensemble-variance (RF) – the ∼16.5× scale gap is visually obvious, the two histograms barely overlap.

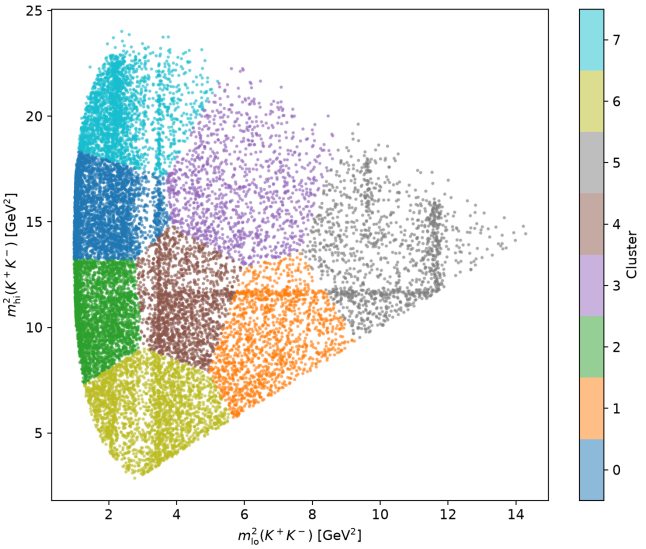


FIG. XI. K-means partition of the Dalitz plane ($K = 8$), same $m^2_{\rm lo}/m^2_{\rm hi}$ axes as Fig. VI. Cluster boundaries follow the point density adaptively, unlike the fixed square grid.

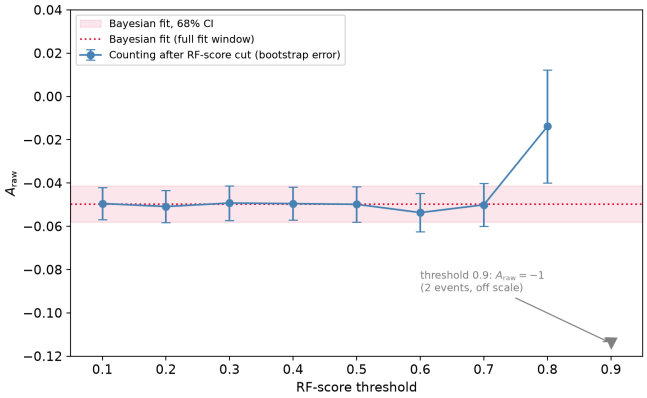


FIG. XII. $A_{\rm raw}$±bootstrap uncertainty vs. RF-score threshold, against the nominal Bayesian fit result (dotted line, shaded 68% CI). Stable through threshold 0.7; at threshold 0.9 only 2 events survive and $A_{\rm raw} = -1$ (marked off scale).

VII. LIMITATIONS AND FUTURE WORK

Several modeling choices and known weaknesses limit how far the results of Secs. IV–V can be pushed, and are collected here rather than scattered through the results sections.

On the Bayesian side (Sec. IV): the background is assumed to split equally between B charges, while the sideband asymmetry (-0.047 ± 0.029) and the full-window count both point to a background asymmetry of similar size to the signal one (Sec. IV F) – the estimated impact on $A_{\rm raw}$ is small, but a fit with independent $N_b^{\pm}$ is needed to settle it; the Savage-Dickey Bayes factor of Sec. IV D is known only approximately ($BF_{01} \sim 10^{-6}$ from the Gaussian approximation, KDE estimates unusable this far in the tail) and depends on the prior width, though the conclusion (decisive evidence against $A = 0$) is robust; the toy-MC coverage check of Sec. IV G is a closure test and cannot detect model misspecification; even the best-fitting signal model (Crystal Ball with floated tail parameters) leaves $\chi^2/\text{ndof} \approx 2$, not ≈ 1 (Sec. IV E); and the fixed-binning Dalitz analysis of Sec. IV H uses a coarse 8×8 grid and a simple mass-window cut rather than a proper background subtraction.

On the machine-learning side (Sec. V): the weak labels suffer from severe class imbalance (1217 background-like vs. 15,119 signal-like events, Sec. V A); the two epistemic-uncertainty estimators (MC Dropout, RF ensemble-variance) disagree by a factor ∼16.5 in absolute scale and are only moderately correlated event-by-event ($\rho \approx 0.37$, Sec. V D); the GBM and FFNN hyperparameter/architecture searches (Tables V–VI) show a winner's-

curse effect (best-by-CV configuration underperforms the untuned default on the test set) that widening the grid further would only worsen, not fix; and the A_{raw}-vs-threshold stability check (Sec. V F) was originally simple post-cut counting only – since addressed on the Bayesian side with a real mass refit on the ML-selected sample (Sec. V F), but the equivalent frequentist-side refit still depends on Sec. III.

Cross-cutting the whole project: the mass window, background shape, and signal-region definition were fixed from the Bayesian fit alone and still need to be reconciled against Marco's frequentist choices before the comparison of Sec. VI can be considered final; and no detector-level systematic correction (selection efficiency, production asymmetry) is applied anywhere in this work, so A_{raw} as reported throughout is explicitly *not* a measurement of A_{CP} (Sec. VI).

The most natural next steps, in rough priority order, are: replacing the fixed-cut local Dalitz asymmetries (both binned, Sec. IV H, and K-means, Sec. V E) with a proper sWeight-based background subtraction; and running the equivalent frequentist-side post-selection mass refit on the ML-selected sample (the Bayesian-side refit is now done, Sec. V F).

VIII. CONCLUSIONS

[TODO: Shared. Short – synthesize ONE key message per method (does not need to restate every number from the Results sections), following the reference report's style: state what was learned methodologically (e.g. "a minimal Gaussian+exponential model is insufficient, a Crystal Ball with floated shape parameters improves the fit without moving A_{raw}"), not just what was measured.]

IX. AUTHOR CONTRIBUTIONS

Marco Belotti led the frequentist analysis (Sec. III): completion of the baseline notebook, mass model, and profile-likelihood confidence interval. Giacomo Moretti led the Bayesian analysis (Sec. IV): model definition, MCMC sampling, convergence diagnostics, Bayes factor, and the associated systematic/goodness-of-fit studies. The machine learning module (Sec. V), the phase-space/toy-MC validation work, the Dalitz-plot analyses, and the final cross-method comparison (Sec. VI) were carried out jointly. Both authors contributed to the manuscript.

X. CODE AND DATA AVAILABILITY

All code is written in Python ≥3.12, run in a dedicated `conda` environment (`cern`). Key package versions: `uproot 5.7`, `iminuit 2.32` **[TODO: Marco confirm]**, `emcee 3.1.6`, `corner 2.3.0`, `scikit-learn 1.9`, `tensorflow 2.21.0`, `keras 3.15.1`, `scipy 1.18`, `numpy 2.4`, `pandas 3.0`.

The dataset is the CERN Open Data record 4900 ("Matter Antimatter Differences – B meson decays to three hadrons"). Analysis code is organized per branch in the project git repository: `Code_general/Frequentist/`, `Code_general/Bayesian/`, `Code_general/Machine_Learning/`.

Appendix A: Full results tables

[TODO: Shared. Still needed: frequentist fit results (Table, once Marco has them) alongside the Bayesian nominal row below, for a direct side-by-side comparison.]

Both Tables V and VI illustrate the same effect for GBM and the FFNN: the configuration with the best *cross-validated* score is not the one with the best *test-set* score, and in both cases test performance is worse than the untuned default. This is not a bug but a known statistical phenomenon – the "winner's curse" of hyperparameter selection: picking the arg-max over many (168 for the FFNN, 100–150 per model here) noisy cross-validation estimates systematically overestimates that configuration's true generalization performance, since the selection itself is biased toward whichever configuration got a favorable CV fluctuation. Searching an even wider grid would make this effect *worse*, not better, since it widens the pool of candidates one of which can get "lucky" on CV – the correct response is to report the CV/test gap honestly (as done here) rather than to keep expanding the search, and to treat the reported "best" configuration's absolute performance with some skepticism even though the underlying search procedure is not at fault. Random Forest does not show this effect (CV and test both improve together), consistent with the well-known property that Random Forest does not overfit as more trees are added – unlike a hyperparameter grid search, which *can* overfit to the validation metric given enough candidates.

[1] R. Aaij *et al.* (LHCb), Direct CP violation in charmless three-body decays of $B^{\pm}$ mesons, Phys. Rev. D **108**, 012008 (2023), arXiv:2206.07622 [hep-ex].

[2] L. Falcão, CP violation measurements in three-body charmless B decays, on behalf of the LHCb Collaboration, 12th Workshop on the CKM Unitarity Triangle (2024),

Variant	n	A_{raw} (median)	68% CI
Nominal ([5150, 5450], exp. bkg)	17839	-0.0497	$[-0.0579, -0.0413]$
Narrower window ([5200, 5450])	17267	-0.0497	$[-0.0580, -0.0413]$
Shorter high-mass tail ([5150, 5400])	17590	-0.0498	$[-0.0581, -0.0415]$
Linear background	17839	-0.0498	$[-0.0581, -0.0415]$
Narrow prior, $A \in [-0.3, 0.3]$	17839	-0.0495	$[-0.0577, -0.0414]$

TABLE III. Bayesian systematic-variation study (Sec. IV F): A_{raw} under three alternative modeling choices and a narrower prior on A, each refit with the same chain length as the nominal fit (9000 steps, 32 walkers). The five medians span 3×10^{-4}, about 4% of the statistical uncertainty.

Model	Test AUC	Brier score	Notes
Logistic Regression	0.7326	0.208	
Logistic Regression (L1)	0.7333	0.212	
Decision Tree	0.7575	0.153	
Random Forest	0.8127	0.132	AUC tied
Gradient Boosting	0.8127	0.125	AUC tied
FFNN	0.7681	0.071	best Brier

TABLE IV. Full model comparison (20-feature classifier), test-set AUC and (uncalibrated) Brier score, all six classifiers of Sec. V B. Isotonic recalibration of the Random Forest score reduces its Brier score to 0.053 (Sec. V C).

arXiv:2403.19921 [hep-ex].
[3] LHCb Collaboration, Matter Antimatter Differences – B meson decays to three hadrons (2017), record 4900.
[4] LHCb Collaboration, Matter Antimatter Differences – Project Notebook (2017), record 4902.

Model	Best configuration (repeated-CV grid search)	CV AUC / Test AUC
Random Forest	`max_depth`= 12, `min_samples_leaf`= 10, `n_estimators`= 1200	0.8172 / **0.8212**
Gradient Boosting	`learning_rate`= 0.05, `max_depth`= 1, `n_estimators`= 500	0.8183 / 0.8066

TABLE V. Hyperparameter search results (`RepeatedStratifiedKFold` CV; Sec. VII), 20-feature classifier, after extending the grid along the three dimensions whose optimum first landed on a grid boundary (Random Forest `n_estimators`, GBM `max_depth`, and the FFNN learning rate of Table VI). Random Forest's optimum is still at the new upper boundary (1200) – diminishing returns, and not the model actually used downstream (Secs. V D–V F always use the untuned default for consistency), so not extended further. GBM's optimum confirms the shallow-tree hypothesis (`max_depth`= 1, i.e. decision stumps) that motivated testing it, but its CV AUC gain (0.8145 → 0.8183) does *not* carry over to test AUC, which is instead worse than both the earlier, narrower-grid result and the untuned default of Table II – see the discussion below.

Architecture	Layers	Dropout	Learning rate	CV AUC
`wide_2layer` (best)	[64, 32]	0.1	0.003	0.7701 ± 0.0209
`deep_5layer`	[128, 64, 32, 16, 8]	0.4	0.005	0.7698 ± 0.0218
`deep_4layer`	[128, 64, 32, 16]	0.1	0.005	0.7685 ± 0.0222
`deep_3layer`	[64, 32, 16]	0.1	0.003	0.7677 ± 0.0204
`deep_4layer`	[128, 64, 32, 16]	0.4	0.005	0.7676 ± 0.0177

TABLE VI. Top-5 configurations from the FFNN architecture search (7 architectures × 4 dropout rates × 6 learning rates = 168 configurations after extending the learning-rate grid to $[0.0002, \ldots, 0.005]$, each 10-fold repeated-CV). The learning-rate boundary effect seen with the narrower grid (top configurations clustering at the old maximum, 0.002) is largely resolved here – the top 5 now split between 0.003 and the new maximum 0.005, consistent with approaching a plateau rather than an unbounded trend. However, the best-by-CV configuration's *test* AUC (0.7665) is actually slightly *below* the untuned fixed-architecture default (0.7675, Table II) – see the discussion below.